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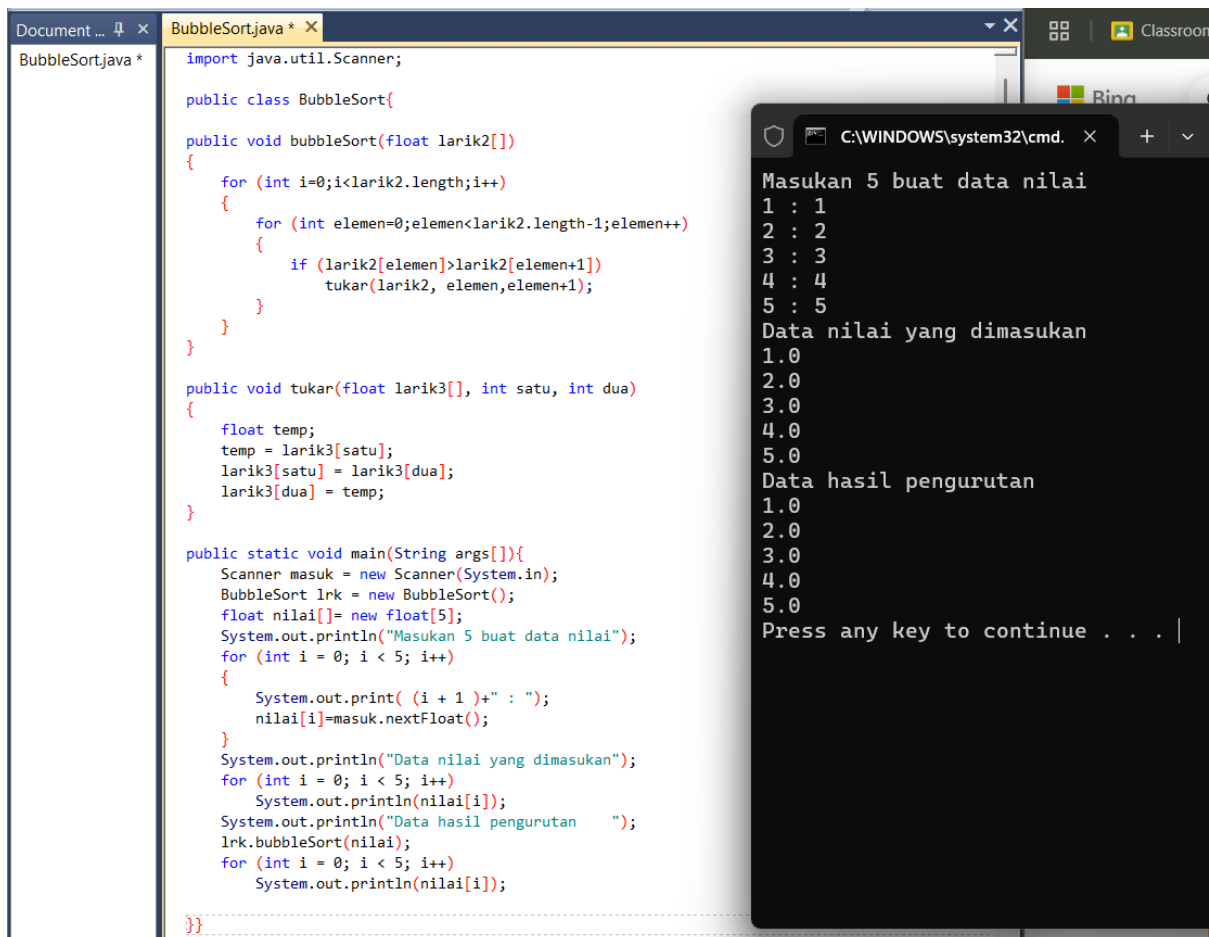
LISTING PRAKTIKUM ALGORITMA DAN PEMROGRAMAN LANJUT

MODUL 13

SORTING

PRAKTIK

Praktik 1



```
Document ... x BubbleSort.java * x
BubbleSort.java *
import java.util.Scanner;

public class BubbleSort{

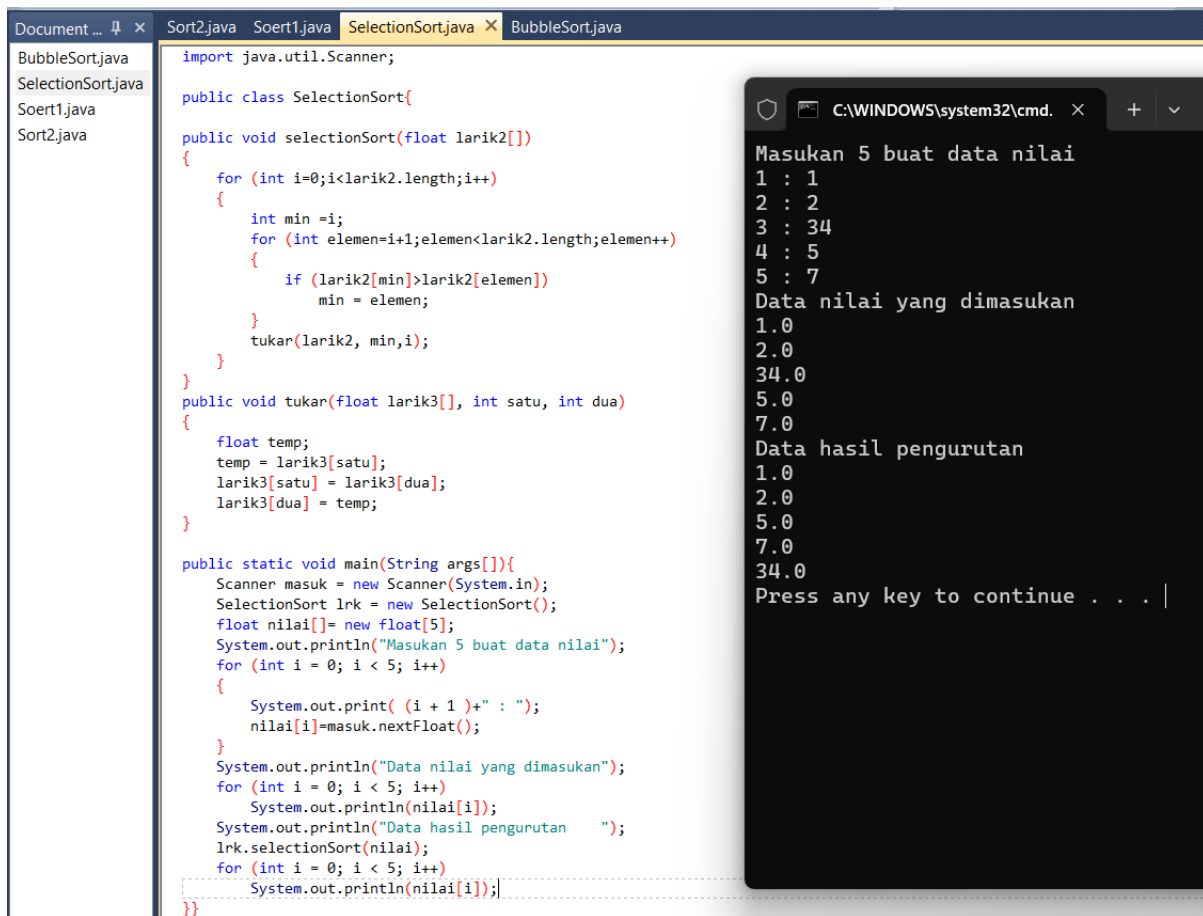
public void bubbleSort(float larik2[])
{
    for (int i=0;i<larik2.length;i++)
    {
        for (int elemen=0;elemen<larik2.length-1;elemen++)
        {
            if (larik2[elemen]>larik2[elemen+1])
                tukar(larik2, elemen,elemen+1);
        }
    }
}

public void tukar(float larik3[], int satu, int dua)
{
    float temp;
    temp = larik3[satu];
    larik3[satu] = larik3[dua];
    larik3[dua] = temp;
}

public static void main(String args[]){
    Scanner masuk = new Scanner(System.in);
    BubbleSort lrk = new BubbleSort();
    float nilai[] = new float[5];
    System.out.println("Masukan 5 buat data nilai");
    for (int i = 0; i < 5; i++)
    {
        System.out.print( (i + 1) + " : ");
        nilai[i]=masuk.nextFloat();
    }
    System.out.println("Data nilai yang dimasukan");
    for (int i = 0; i < 5; i++)
        System.out.println(nilai[i]);
    System.out.println("Data hasil pengurutan");
    lrk.bubbleSort(nilai);
    for (int i = 0; i < 5; i++)
        System.out.println(nilai[i]);
}
}
```

```
C:\WINDOWS\system32\cmd. x
Masukan 5 buat data nilai
1 : 1
2 : 2
3 : 3
4 : 4
5 : 5
Data nilai yang dimasukan
1.0
2.0
3.0
4.0
5.0
Data hasil pengurutan
1.0
2.0
3.0
4.0
5.0
Press any key to continue . . . |
```

Praktik 2



The screenshot shows an IDE with four tabs: Sort2.java, Soert1.java, SelectionSort.java (active), and BubbleSort.java. The SelectionSort.java file contains the following code:

```
import java.util.Scanner;

public class SelectionSort{

public void selectionSort(float larik2[])
{
    for (int i=0;i<larik2.length;i++)
    {
        int min =i;
        for (int elemen=i+1;elemen<larik2.length;elemen++)
        {
            if (larik2[min]>larik2[elemen])
                min = elemen;
        }
        tukar(larik2, min,i);
    }
}

public void tukar(float larik3[], int satu, int dua)
{
    float temp;
    temp = larik3[satu];
    larik3[satu] = larik3[dua];
    larik3[dua] = temp;
}

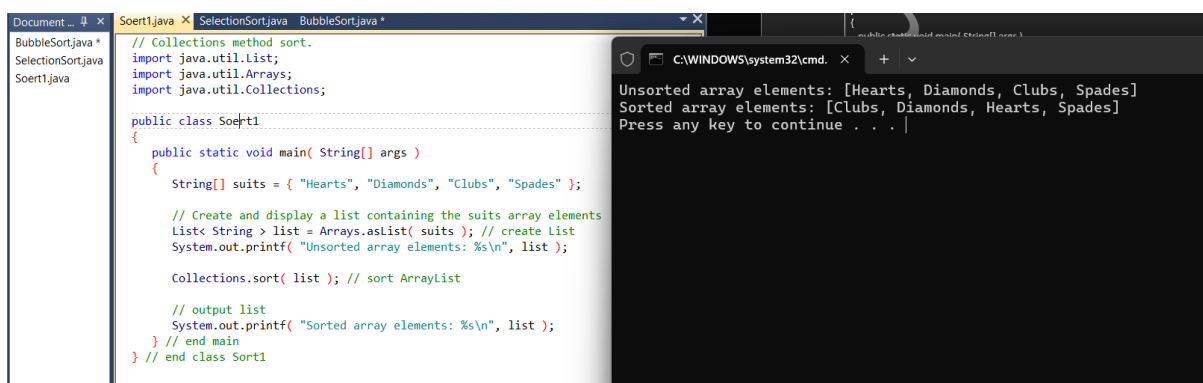
public static void main(String args[]){
    Scanner masuk = new Scanner(System.in);
    SelectionSort lrk = new SelectionSort();
    float nilai[] = new float[5];
    System.out.println("Masukan 5 buat data nilai");
    for (int i = 0; i < 5; i++)
    {
        System.out.print( (i + 1) + " : ");
        nilai[i]=masuk.nextFloat();
    }
    System.out.println("Data nilai yang dimasukan");
    for (int i = 0; i < 5; i++)
        System.out.println(nilai[i]);
    System.out.println("Data hasil pengurutan ");
    lrk.selectionSort(nilai);
    for (int i = 0; i < 5; i++)
        System.out.println(nilai[i]);
}}
```

The command prompt output shows the execution of the program:

```
C:\WINDOWS\system32\cmd. x + v
Masukan 5 buat data nilai
1 : 1
2 : 2
3 : 34
4 : 5
5 : 7
Data nilai yang dimasukan
1.0
2.0
34.0
5.0
7.0
Data hasil pengurutan
1.0
2.0
5.0
7.0
34.0
Press any key to continue . . . |
```

LATIHAN

Latihan 1



The screenshot shows an IDE with three tabs: Soert1.java (active), SelectionSort.java, and BubbleSort.java. The Soert1.java file contains the following code:

```
// Collections method sort.
import java.util.List;
import java.util.Arrays;
import java.util.Collections;

public class Soert1
{
    public static void main( String[] args )
    {
        String[] suits = { "Hearts", "Diamonds", "Clubs", "Spades" };

        // Create and display a list containing the suits array elements
        List< String > list = Arrays.asList( suits ); // create List
        System.out.printf( "Unsorted array elements: %s\n", list );

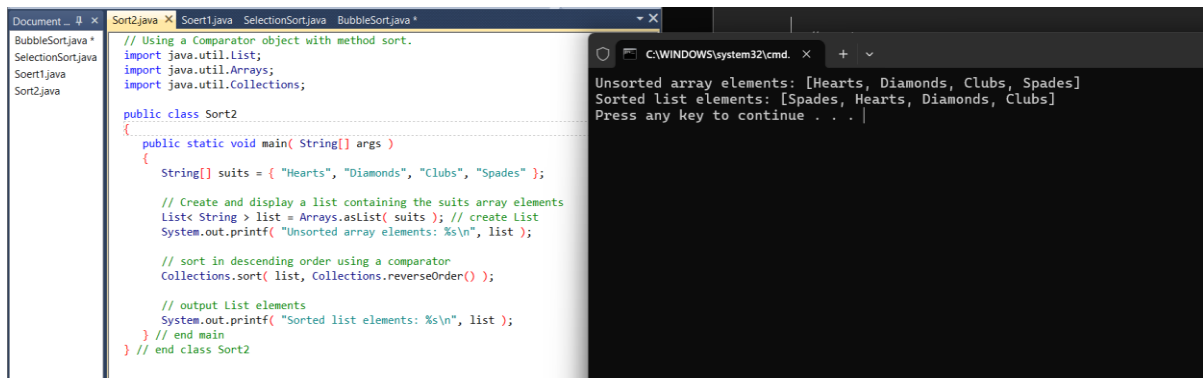
        Collections.sort( list ); // sort ArrayList

        // output list
        System.out.printf( "Sorted array elements: %s\n", list );
    } // end main
} // end class Soert1
```

The command prompt output shows the execution of the program:

```
C:\WINDOWS\system32\cmd. x + v
Unsorted array elements: [Hearts, Diamonds, Clubs, Spades]
Sorted array elements: [Clubs, Diamonds, Hearts, Spades]
Press any key to continue . . . |
```

Latihan 2



The screenshot shows a Java IDE with several files open: `BubbleSort.java`, `SelectionSort.java`, `Sort1.java`, and `Sort2.java`. The `Sort2.java` file is the active one, displaying the following code:

```
// Using a Comparator object with method sort.
import java.util.List;
import java.util.Arrays;
import java.util.Collections;

public class Sort2
{
    public static void main( String[] args )
    {
        String[] suits = { "Hearts", "Diamonds", "Clubs", "Spades" };

        // Create and display a list containing the suits array elements
        List< String > list = Arrays.asList( suits ); // create List
        System.out.printf( "Unsorted array elements: %s\n", list );

        // sort in descending order using a comparator
        Collections.sort( list, Collections.reverseOrder() );

        // output list elements
        System.out.printf( "Sorted list elements: %s\n", list );
    } // end main
} // end class Sort2
```

To the right of the IDE is a Windows command prompt window titled `C:\WINDOWS\system32\cmd`. It displays the output of the program:

```
Unsorted array elements: [Hearts, Diamonds, Clubs, Spades]
Sorted list elements: [Spades, Hearts, Diamonds, Clubs]
Press any key to continue . . . |
```

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TUGAS

Tugas 1

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